REPORT DOCUMENTATION PAGE

Form Approved
OMB No. 0704-0188

The public reporting burden for this collection of information is estimated to average 1 hour per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden, to Department of Defense, Washington Headquarters Services, Directorate for information on Operations and Reports (0704-0188), 1215 Jefferson Davis Highway, Suite 1204, Arlington, VA 22202-4302. Respondents should be aware that notwithstanding any other provision of law, no person shall be subject to any penalty for failing to comply with a collection of information if it does not display a currently valid OMB control number.
PLEASE DO NOT RETURN YOUR FORM TO THE ABOVE ADDRESS.

1. REPORT DATE *(DD-MM-YYYY)*	2. REPORT TYPE	3. DATES COVERED *(From - To)*
19-07-2016	Final	

4. TITLE AND SUBTITLE
Test Operations Procedure (TOP)
06-2-600 Multiple-Input Multiple-Output Antenna System Measurements

5a. CONTRACT NUMBER

5b. GRANT NUMBER

5c. PROGRAM ELEMENT NUMBER

6. AUTHORS

5d. PROJECT NUMBER

5e. TASK NUMBER

5f. WORK UNIT NUMBER

7. PERFORMING ORGANIZATION NAME(S) AND ADDRESS(ES)
U.S. Army Electronic Proving Ground
2000 Arizona Street
Fort Huachuca, AZ 85613-7063

8. PERFORMING ORGANIZATION REPORT NUMBER
TOP 06-2-600

9. SPONSORING/MONITORING AGENCY NAME(S) AND ADDRESS(ES)
Range Infrastructure Division (CSTE-TM)
U.S. Army Test and Evaluation Command
2202 Aberdeen Boulevard
Aberdeen Proving Ground, MD 21005-5001

10. SPONSOR/MONITOR'S ACRONYM(S)

11. SPONSOR/MONITOR'S REPORT NUMBER(S)
Same as item 8

12. DISTRIBUTION/AVAILABILITY STATEMENT

Distribution Statement A. Approved for public release; distribution is unlimited.

13. SUPPLEMENTARY NOTES
Defense Technical Information Center (DTIC), AD No.: AD1012096

14. ABSTRACT
This Test Operations Procedure (TOP) gives guidelines for the testing of multiple-input multiple-output (MIMO) antenna systems during development testing (DT) by the U.S. Army Test and Evaluation Command (ATEC).

15. SUBJECT TERMS
MIMO, antenna characterization, antenna pattern

16. SECURITY CLASSIFICATION OF:			17. LIMITATION OF ABSTRACT	18. NUMBER OF PAGES	19a. NAME OF RESPONSIBLE PERSON
a. REPORT	B. ABSTRACT	C. THIS PAGE			19b. TELEPHONE NUMBER *(include area code)*
Unclassified	Unclassified	Unclassified	SAR	20	

Standard Form 298 (Rev. 8-98)
Prescribed by ANSI Std. Z39-18

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

(This page is intentionally blank.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

US ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

*Test Operations Procedure 06-2-600 19 July 2016
DTIC AD No. AD1012096

MULTIPLE-INPUT MULTIPLE-OUTPUT ANTENNA SYSTEM MEASUREMENTS

			Page
Paragraph	1.	SCOPE	2
	2.	FACILITIES AND INSTRUMENTATION	2
	2.1	Facilities	2
	2.2	Instrumentation	3
	3.	SAFETY TESTING	3
	3.1	General	3
	3.2	Measurement and Tolerances	4
	4.	REQUIRED TEST CONDITIONS	5
	5.	TEST PROCEDURES	6
	5.1	Figures of Merit	6
	5.2	Suggested Equipment	7
	5.3	Test Methodologies	8
	6.	DATA REQUIRED	10
	6.1	Common Data Required	10
	6.2	Test Data Required	10
	7.	PRESENTATION OF DATA	11
APPENDIX	A.	ABBREVIATIONS	A-1
	B.	REFERENCES	B-1
	C.	APPROVAL AUTHORITY	C-1

Approved for public release; distribution unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

1. SCOPE.

a. This Test Operations Procedure (TOP) gives guidelines for the testing of multiple-input multiple-output (MIMO) antenna systems. The specific and complicated nature of MIMO systems requires a large degree of flexibility in test setup, execution, and analysis. As with software, each implementation of MIMO is supported with a unique set of commands, and executed with its own logic. It is assumed throughout this document that the command set, or software behind the MIMO antennas under test (AUTs), will stay outside the purview of immediate analysis, and will remain a control. This document assumes multipath propagation to be defined as the arrival of radio signals at a defined location by two or more paths. The document also requires over-the-air (OTA) testing as the basis for analyzing the systems. Two anechoic chamber-based testing methods are suggested, one using a channel emulator to simulate traffic and multipath, and the other using a multipath simulation in a controlled environment for situations when a channel emulator is not available or necessary (a common occurrence).

b. This TOP is applicable for complete systems and platforms, both new and modified, and applies to engineering development, production, and sustainment phases of the system. It should be noted that the Man/Soldier is a designated platform and will be addressed in that manner for safety and performance requirements.

c. The results of the tests conducted as outlined in this procedure will enable an analysis to be made as to whether the system or platform tested meets the pre-agreed requirements when subjected to the defined conditions.

d. The procedure applies to all hardware- and software-realized MIMO antenna configurations and systems.

2. FACILITIES AND INSTRUMENTATION.

2.1 Facilities.

a. The facility must be capable of handling the test item in all of its configurations, from storage to pre-launch/operational and in-flight configurations as applicable; specifically stockpile to safe launch (for ordnance items) and in-flight (for guided ordnance).

b. Fabrication facilities are required where instrumentation system(s) can be fabricated and installed in the test item in a manner that will not affect the antenna characteristics of the tactical configuration of the test item, and that will protect the instrumentation from electromagnetic environmental effects (E3) and signal degradation. Instrumentation should consist of sensing elements, transmitting units, and associated interconnections. Appropriate channel models should be provided when available, and are a necessary provision if they are a required test element. The equipment required in the fabrication facilities includes such items as machine tools, micromanipulators, sheet metal fabricating tools, microscopes, pneumatic switching, and equipment associated with fiber-optic technology.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

2.2 Instrumentation.

a. All test equipment must be calibrated annually by calibration laboratories using procedures traceable to the National Institute of Standards and Technology (NIST).

b. System under test (SUT) instrumentation is permissible but it shall not influence test results. To minimize the possibility of instrumentation influencing test results or instrumentation susceptibility, the following will apply:

(1) Use fiber-optic based instrumentation whenever possible.

(2) Use twisted, shielded pair wiring when conductive leads are required.

(3) Instrumentation equipment and its leads should be buried inside the SUT with only the fiber-optic leads exiting the SUT, whenever possible.

(4) Antennas should be mounted as deployed, with no wiring or extraneous equipment altering normal patterns of transmission.

3. SAFETY TESTING.

3.1 General.

a. When the SUT contains ordnance or safety-critical circuits and functions [e.g., electrically initiated devices (EIDs) (previously called electro-explosive devices (EEDs)], the SUT ordnance and safety-critical circuits/functions shall meet their safety requirements during and after exposure to the electromagnetic environment. The following methods are routinely used on Army materiel; additional guidance is provided in Military Handbook (MIL-HDBK)-240A[1**]. As all suggested procedures utilize electromagnetic radiation, Joint Ordnance Test Procedure (JOTP)-061[2] shall be used for guidance in testing of Joint munitions to hazards of electromagnetic radiation to ordnance (HERO). A second JOTP is in process and shall be used for guidance of Joint electrostatic discharge (ESD) and lightning testing when complete.

b. In accordance with JOTP-061 and MIL-HDBK-240A, care shall be exercised when performing HERO testing to ensure that personnel are not exposed to radio frequency (RF) electromagnetic field levels that exceed the appropriate exposure reference levels specified in the most current Department of Defense Instruction (DODI) 6055.11[3]. Each test operation should be evaluated for compliance with the RF exposure guidelines to determine if an RF safety program is necessary. During all phases of testing, ensure that appropriate RF safety program elements are in place such as the placement of appropriate RF safety signage, designation of restricted access areas, and RF safety awareness training.

** Superscript numbers correspond to Appendix B, References.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

c. There are two methods for testing EID safety, which are described in detail in Section 3.6 (HERO Test Philosophy) of Army Technical Report (TR)-RD-TE-97-01[4]. Electromagnetic radiation hazards (EMRH) is an antiquated term for HERO. The two methods for testing safety of SUT EIDs are:

(1) Instrumented, inert SUT Testing. The SUT EIDs are replaced by inert versions (explosive material removed for personnel and SUT safety) that are instrumented to measure the currents, voltages, power, or energy induced at the EIDs during safety testing. The measured results are compared to the EID maximum no-fire (MNF) parameter after the MNF parameter is reduced by the EID-required safety margin. Note that the measurement results should be extrapolated if the test facility cannot produce the required test levels (refer to MIL-HDBK-240A).

(2) Go/No-Go SUT Testing. The SUT EIDs are left intact in the SUT and are not instrumented. A large number of SUTs with their live EIDs are tested to the test level in order to obtain a meaningful statistical sample. If a large number of SUTs or EIDs are not available, a smaller number of SUTs are subjected to test levels that are raised by the required EID safety margin. The justification for the chosen sample size, along with assumptions, the statistical test used, confidence levels (Army's risk), and power (Manufacturer's risk) shall be addressed in the test plan. Testing repeated 10 times with no failures would give 80-percent confidence in the SUT having at least an 80-percent success rate (Clopper-Pearson method). Under the same method, a test with 22 repetitions and no failures would give a 90-percent confidence in the system having at least a 90-percent success rate. A higher number of repetitions should be used for airborne systems based on the additional risk in flight. Fresh EIDs should be used for each test run to provide some measure of confidence that the EIDs will remain safe.

3.2 Measurement and Tolerances.

Unless otherwise stated for a particular measurement, the following historical tolerances must be followed:

Item	Requirement
Distance	± 5 percent
Frequency	± 2 percent
Amplitude, measurement receiver	± 2 decibels (dB)
Amplitude measurement system (includes measurement receivers, transducers, cables, etc.)	± 3 dB

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

Item	Requirement
Inert EID instrumentation	The measured voltage output of the thermocouple, vacuum thermocouple, or optical temperature sensor shall indicate, within ± 2 percent, the actual EID stimulus.
Sensitivity	< 5 percent of the EID maximum no-fire current (MNFC), applicable where MNFC is above 100 milliamperes (mA).
Sensors/detectors	The measured output of the sensors/detectors shall indicate, within ± 2 percent, the variable being measured.
Fiber-optics data link	± 5 percent of the measured variable.
Current meter/monitor	± 10 percent of actual discharge current.
Magnetic field probe	± 10 percent of the actual strike magnetic field.
Field intensity probe	± 2 dB of actual electric field.
Waveform receiver	± 5 percent of maximum peak discharge current.
Field intensity meter	± 2 dB of actual field intensity.
Voltage meter/monitor	± 10 percent of actual charge voltage
EID tester	± 10 percent of the full-scale meter setting.

4. REQUIRED TEST CONDITIONS.

Test item considerations prior to the start of the test include but are not limited to:

a. SUT description, components, manufacturer, part number (PN) and serial number (SN), etc., including specific power source requirements.

b. Human interaction, mounted/worn equipment.

c. System interfaces, including access to antenna ports and the ability to access an individual antenna in a MIMO system.

d. Description of built-in SUT electromagnetic hardening protection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

e. Test environments and limits.

f. Verification issues.

g. Pass/fail criteria, performance specifications, and margins required.

h. Test configurations and modes of operation and SUT control settings.

i. Peripheral equipment required (e.g., test fixture, test set, power supply).

j. Description of actions, outputs, or displays monitored.

k. Security and classification requirements for data and results.

l. EID electrical parameters data sheets for HERO calibration and data collection.

m. Safety Assessment Report (SAR).

5. TEST PROCEDURES.

5.1 Figures of Merit.

a. The primary figure of merit in testing MIMO antenna systems used for digital networks is throughput, where throughput is the time-averaged number of correctly received transport blocks in a communication system running an application.

(1) In this case, the transport block is defined as a single unit in the reference measurement channel.

(2) Throughput is determined as: Throughput = [Transmitted Transport Block Size x Number of Acknowledged Signals (ACKs)]/Time, where time in this case is defined as a period sufficiently long to average out variations due to the fading channel.

b. Additional figures of merit include:

(1) Gain imbalance.

(2) Spatial correlation.

(3) Antenna efficiency.

(4) Total radiated power (TRP) and total radiated sensitivity (TRS).

(5) Channel quality indicators (CQIs), block error rates (BLERs).

(6) MIMO channel capacity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

5.2 Suggested Equipment.

Suggested equipment is dependent upon the methodology used during test. Two approaches are outlined below, each requiring an anechoic chamber, positioner, test probes, signal source, data recording device, and the SUT.

a. Anechoic Chamber. An anechoic chamber should adequately reduce unwanted reflections from the test environment, and adequately shield the test from unwanted outside radiation. The chamber should achieve ≥ 80 dB attenuation of outside leakage and interference, across the band of operation of the SUT. Additionally, the chamber should ideally be large enough to achieve measurement at a distance outside the near-field distance of the band of operation of the SUT. The near-field distance is:

$$d = \frac{2D^2}{\lambda}$$

where:

D = the maximum dimension of the antenna

λ = the wavelength under test

If a far-field operation is not possible, near-field correction software may be used; however, as of this time it greatly increases the time required for testing and ideally should be avoided.

b. Positioner. A remotely controlled positioner capable of spherical measurements is necessary to achieve a full three-dimensional (3D) pattern measurement.

c. Test Probes. Test probes shall be chosen according to the band of operation of the SUT and should be chosen to minimize reflections in the anechoic chamber. Probes often take the form of common measurement antennas that are easily moved around the chamber according to the needs of the test.

d. Signal Source. The signal source for testing should be capable of emulating multiple channels, or capable of handling separate emulation equipment and software as needed for each specific SUT, across its band of operation, and within established tolerances required for the test. Companies such as Keysight*** produce a variety of integrated vector network analyzers with these built-in functions. Measurement of the system requires the equipment to fully support amplitude and relative phase measurements of the antennas.

*** The use of brand names does not constitute endorsement by the Army or any other agency of the Federal Government, nor does it imply that it is best suited for its intended application.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

e. Data Recording Device. A device, most commonly a computer, shall be used to accurately record required data from the test equipment, according to proper classification procedures. This equipment should not interfere with measurement activities or introduce RF energy to the test.

f. SUT. As outlined in paragraph 4, the SUT must accurately represent the system's state and use for the required data point. Many MIMO methods differ in implementation; therefore, proper multichannel/signal operation must be emulated or created using the signal or source, with the SUT or with a separate set of equipment.

5.3 Test Methodologies.

The two suggested test methodologies require use of an anechoic chamber, which is the most common, compact testing environment available to testing facilities. The methods each consist of OTA testing using a number of test probes positioned in the chamber with the equipment, transmitting signals with temporal and spatial characteristics for testing MIMO antenna systems.

5.3.1 Method One.

The first method assumes access to the MIMO system's channel model, access to active elements (if existing), and the capability to independently apply it to the device under test (DUT). The method is divided into two stages, first determining the antennas' far-field radiation pattern, radiated power, and efficiency. Stage 2 incorporates the antennas' characteristics into the MIMO system's channel models in order to achieve real-time emulation and proper throughput measurement.

a. Stage 1.

(1) Accurately measuring a MIMO system's antenna pattern requires that far-field measurements fully capture the mutual coupling of the complete array. Measurement of the system also requires the equipment to fully support amplitude and relative phase measurements of the antennas.

(2) Pattern measurements should be made using an anechoic chamber. The DUT is put into a chamber and each antenna element's far zone pattern is measured. Measurement in two orthogonal polarizations using a positioner capable of spherical measurements is necessary to achieve a full 3D far-zone pattern measurement. If the measurement probe cannot be placed in the far field, near-field measurements can be taken using near field correction software.

b. Stage 2.

(1) The second stage assumes access to a channel model and a commercially available multipath simulator. The antenna characteristics measured in Stage 1 are incorporated into the channel model with both MIMO antenna effects and the multipath channel effects

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

emulated using a commercial MIMO channel emulator. The emulator is then connected to the DUT's antenna ports using radio frequency (RF) cables.

(2) By controlling the power settings of the channel emulator and the integrated channel model, the end-to-end throughput with the MIMO antenna radiation influence can be measured. There are two different approaches to combine the antenna patterns with the MIMO channel model:

(a) The first approach is to apply antenna patterns to Ray-based channel models. Ray-based models are capable of supporting arbitrary antenna patterns under predefined channel models in a natural way as described above. If a Ray-based model is specified to be used for a MIMO OTA test, then the channel emulator must be capable of supporting the Ray-based model channel model emulation and of supporting loading measured antenna patterns.

(b) The second approach is to apply antenna patterns to correlation-based channel models. With a correlation matrix calculation method for arbitrary antenna patterns under multipath channel conditions, the correlation matrix and the antenna imbalance can be calculated and then emulated by the channel emulator.

5.3.2 Method Two.

a. The second method for testing MIMO antenna systems uses an RF-controlled spatial fading emulator to reproduce a multipath radio propagation environment using emissions from probes arranged around the DUT.

b. A key feature of this method is that it does not use the sophisticated commercial channel emulator. By using the combination of phase shifters, power dividers, and attenuators, operating in the RF band, a realistic fading channel environment can be emulated. To reduce the influence from the measurement equipment, the receiver, phase shifter, power divider, transmitter, and computer are set outside of the anechoic chamber; although proper in-chamber shielding can be accomplished using anechoic dividers.

c. In this method, all signals are operated and controlled at the RF level. A computer (either a laptop or relatively powerful computer) is used to provide the following:

(1) A graphical user interface (GUI) to set the input parameters, determine the measured parameters to be collected, set the calibration parameters, and set the DUT parameters.

(2) The generation of control signals to manipulate the phase angle of each phase shifter.

(3) The collection of measured raw data obtained via the DUT, and post-processing of the measured raw data to derive the desired figures of merit (i.e., minimum requirements for the DUT).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

(4) The initiation of a base station emulator and the starting of the testing session (by establishing a communication session with DUT).

d. The RF signals transmitted from the base station emulator's antenna connector are fed to a bank of power dividers. Each power divider provides an identical RF signal from each of the output ports. The number of power dividers required is determined by ***N,*** or the number of probes.

e. Each power divider output is then fed to a phase shifter.

f. The phase shifter is used to change the phase of the RF signal according to the parameter setting input to the computer earlier. Note that the control signal from the computer is digital-to-analogue (D/A), converted before being used to control the phase shifter. By controlling the phase of each RF signal, a Rayleigh distributed or other relevant multipath distribution can be obtained. The number of phase shifters required is determined by ***N***.

g. The output of the phase shifters is connected to the antenna probes. The signal from each phase shifter is fed to the vertical and horizontal antennas and radiates toward the DUT. The DUT then measures the signals from each antenna probe and the measurement data are reported back to the computer. The amount of measurement data to be collected can be controlled by the computer by setting the sampling rate.

6. DATA REQUIRED.

6.1 Common Data Required.

a. Record test item nomenclature, SN, technical characteristics, and performance parameters, as applicable.

b. Photograph the test item setup in order to provide visual identification and description.

c. Record the test equipment nomenclature, model numbers, SNs, and calibration dates.

d. Record start and completion dates of major test groups (i.e., ambient, climatic, mechanical).

e. Record description, dates, corrective actions, and analyses for test item failures.

6.2 Test Data Required.

a. Collected data will enable an analysis to be made as to whether the system or platform tested meets the pre-agreed requirements when subjected to the defined conditions, and will be specified accordingly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

b. At a minimum, record position-referenced signal levels of the antenna during operation utilizing modes of interest. Gain data and visual representation of pattern data may be required.

c. Gain measurements must include measurement system and reference antenna data used to determine results.

d. Visual representation of antenna pattern and gain data should include appropriate position and axial information, dependent upon specifics of the system and test.

e. Throughput calculated from the time-averaged number of correctly received transport blocks will be recorded and associated with test procedures and conditions.

7. DATA PRESENTATION.

The data collected will be presented numerically and in tables and graphs. Antenna patterns will include position and signal level data for reference, with detail dependent upon the individual test plans. Details are contained in the individual test procedures. In some cases, it may be prudent to list the data in computer files to allow later access and plotting. Figure 1 shows an example of a frequency versus elevation test plot.

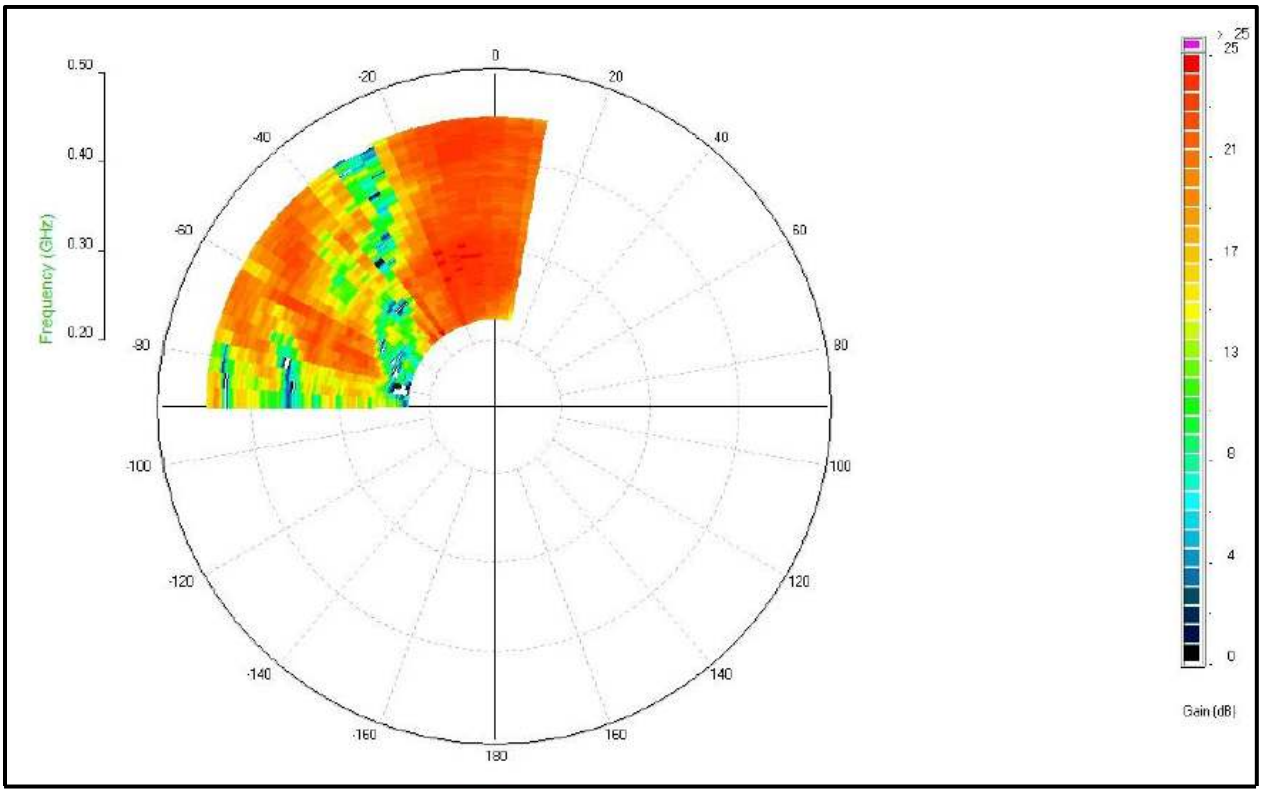


Figure 1. Example plot of frequency versus elevation, 0-degree azimuth.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

(This page is intentionally blank.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

APPENDIX A. ABBREVIATIONS.

3D	three-dimensional
ACK	acknowledged signal
AUT	antenna under test
BLER	block error rate
CQI	channel quality indicator
D/A	digital-to-analog
dB	decibel
DOD	Department of Defense
DODI	Department of Defense Instruction
DUT	device under test
E3	electromagnetic environmental effects
EED	electro-explosive device
EID	electrically initiated device
EMI	electromagnetic interference
EMP	electromagnetic pulse
EMRH	electromagnetic radiation hazard
EMRO	electromagnetic radiation operation(al)
ESD	electrostatic discharge
GUI	graphical user interface
HERO	hazards of electromagnetic radiation to ordnance
JOTP	Joint Ordnance Test Procedure
mA	milliampere
MIL-HDBK	Military Handbook
MIMO	multiple-input multiple-output
MNF	maximum no-fire
MNFC	maximum no-fire current
NIST	National Institute of Standards and Technology
NOSSA	Naval Ordnance Safety and Security Activity
OTA	over-the-air
PN	part number

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

APPENDIX A. ABBREVIATIONS.

RF	radio frequency
SAR	Safety Assessment Report
SN	serial number
SUT	system under test
TOP	Test Operations Procedure
TR	technical report
TRP	total radiated power
TRS	total radiated sensitivity

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

APPENDIX B. REFERENCES.

1. MIL-HDBK-240A, Hazards of Electromagnetic Radiation to Ordnance Test Guide, U.S. Department of Defense (DOD), 10 March 2011.

2. JOTP-061, Hazards of Electromagnetic Radiation to Ordnance (HERO) Safety Test, Naval Ordnance Safety and Security Activity (NOSSA), 1 October 2013.

3. DODI 6055.11, Protecting Personnel from Electromagnetic Fields, 19 August 2009.

4. Army Technical Report (TR)-RD-TE-97-01, Electromagnetic Environmental Effects Criteria and Guidelines for EMRH, EMRO, Lightning Effects, ESD, EMP, and EMI Testing of U.S. Army Missile Systems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

(This page is intentionally blank.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

APPENDIX C. APPROVAL AUTHORITY.

CSTE-TM 19 July 2016

MEMORANDUM FOR

Commanders, All Test Centers
Technical Directors, All Test Centers
Directors, U.S. Army Evaluation Center
Commander, U.S. Army Operational Test Command

SUBJECT: Test Operations Procedure (TOP) 06-2-600 Multiple-Input Multiple-Output Antenna System Measurements, Approved for Publication

1. TOP 06-2-600 Multiple-Input Multiple-Output Antenna System Measurements, has been reviewed by the U.S. Army Test and Evaluation Command (ATEC) Test Centers, the U.S. Army Operational Test Command, and the U.S. Army Evaluation Center. All comments received during the formal coordination period have been adjudicated by the preparing agency. The scope of the document is as follows:

This TOP provides guidelines for the testing of multiple-input multiple-output (MIMO) antenna systems. The specific and complicated nature of MIMO antenna systems requires a large degree of flexibility in test setup, execution, and analysis. It is assumed throughout this document that the command set, or software behind the MIMO antennas under test, will stay outside the purview of immediate analysis, and will remain a control.

2. This document is approved for publication and will be posted to the Reference Library of the ATEC Vision Digital Library System (VDLS). The VDLS website can be accessed at https://vdls.atc.army.mil/.

3. Comments, suggestions, or questions on this document should be addressed to U.S. Army Test and Evaluation Command (CSTE-TM), 2202 Aberdeen Boulevard-Third Floor, Aberdeen Proving Ground, MD 21005-5001; or e-mailed to usarmy.apg.atec.mbx.atec-standards@mail.mil.

CHEW.JENNIFER.PM.1041816836

JENNIFER P. CHEW
Associate Director, Test Management Directorate (G9)

FOR

RAYMOND G. FONTAINE
Director, Test Management Directorate (G9)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

(This page is intentionally blank.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

Forward comments, recommended changes, or any pertinent data which may be of use in improving this publication to the following address: Range Infrastructure Division (CSTE-TM), US Army Test and Evaluation Command, 2202 Aberdeen Boulevard, Aberdeen Proving Ground, Maryland 21005-5001. Technical information may be obtained from the preparing activity: U.S. Army Electronic Proving Ground, 2000 Arizona Street, Fort Huachuca, Arizona 85613-7063. Additional copies can be requested through the following website: http://www.atec.army.mil/publications/topsindex.aspx, or through the Defense Technical Information Center, 8725 John J. Kingman Rd., STE 0944, Fort Belvoir, VA 22060-6218. This document is identified by the accession number (AD No.) printed on the first page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.